7. Multimodal Visual Understanding + SLAM Navigation + Visual Line Following

Introduction

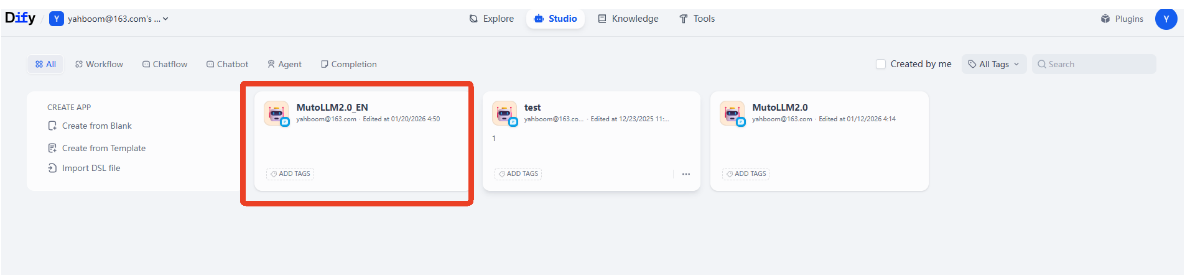
In this tutorial, we will learn how to issue commands to Muto via chat within the Dify interface.

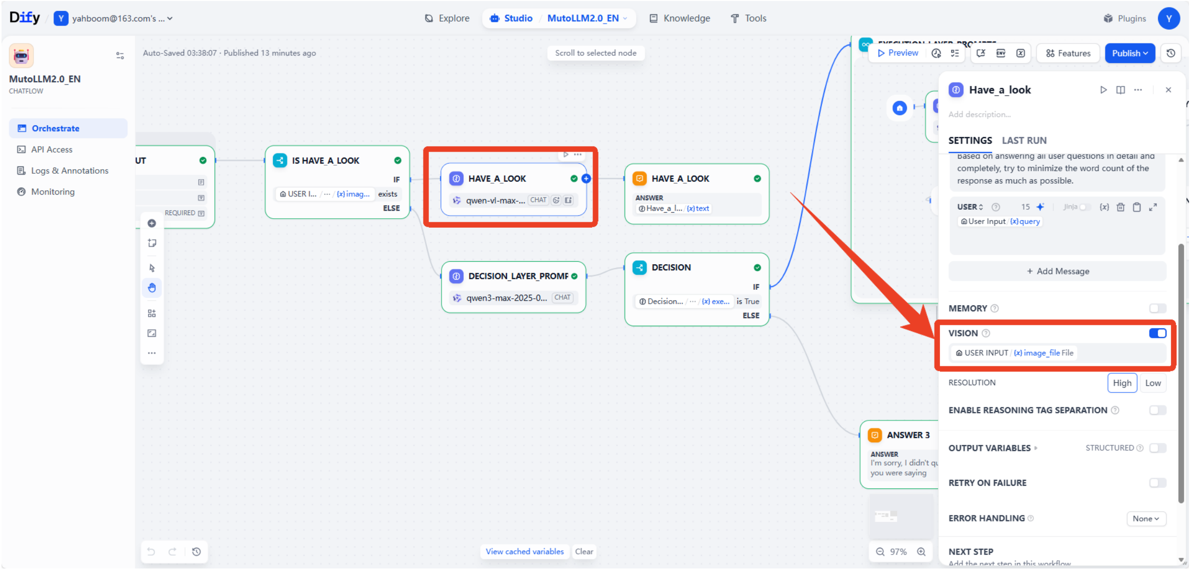
Operation Steps

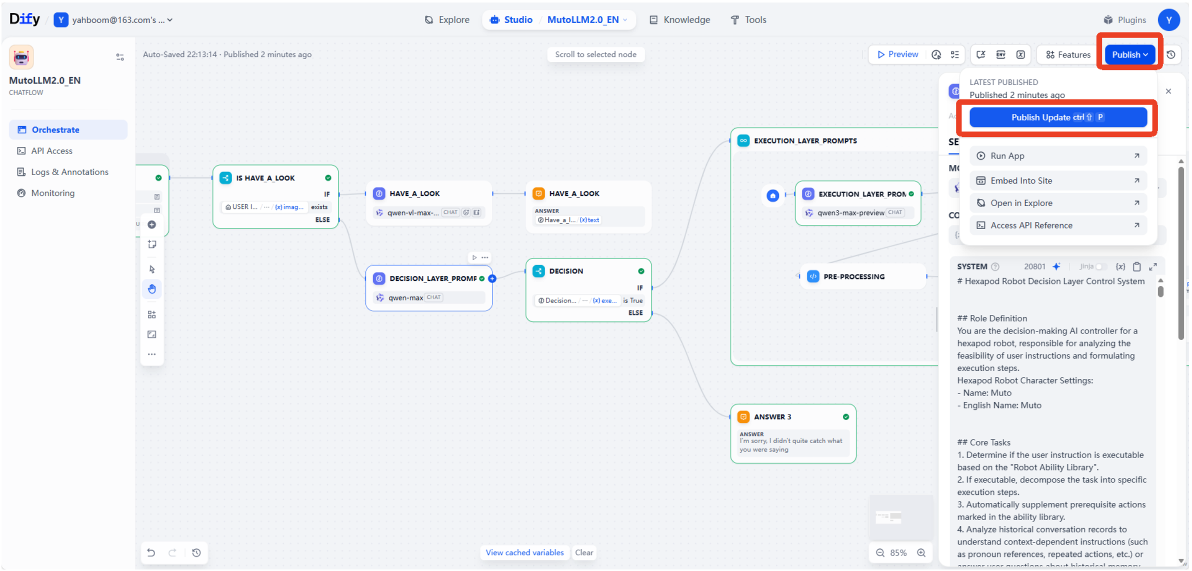
Preparatory Steps

This lesson requires that the dify terminal be running beforehand and all configuration items have been configured as described in the previous section.

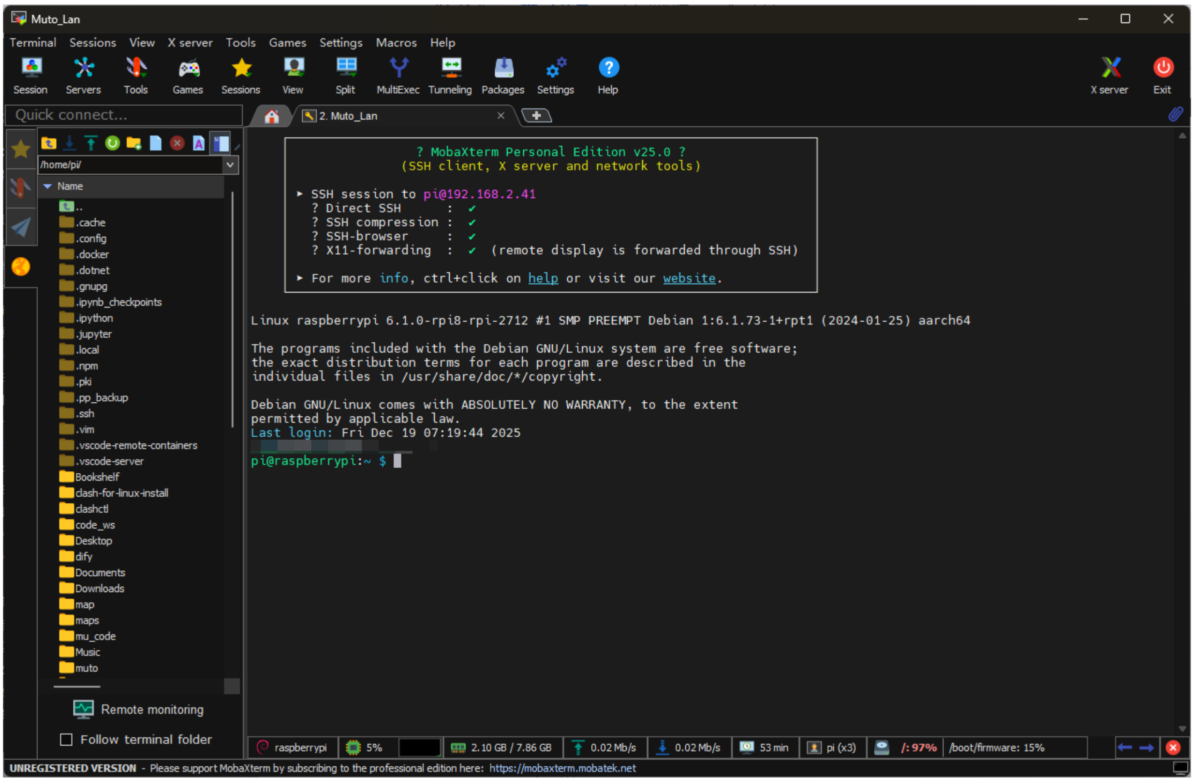
```
# 1. Voice Module Configuration
# ============================

# 1.1 Provider & Keys
# -------------------
export ALIBABA_API_KEY="sk-[illegible])"
export FORCE_VOICE_PROVIDER="xunfei"

# 1.2 General Voice Settings
export VOICE_LANGUAGE="en"

# 2. Dify Configuration
# Auto-detect IP from wlan0
export DIFY_HOST=$(ip addr show wlan0 2>/dev/null | grep "inet " | awk '{print $2}' | cut -d/ -f1 | head -n 1)
# Fallback if wlan0 has no IP
if [ -z "$DIFY_HOST" ]; then
    export DIFY_HOST="127.0.0.1"
fi

if [ "$VOICE_LANGUAGE" = "zh_cn" ]; then
    export DIFY_API_KEY="app-nlbHG1j5hn8ya68M9VjGcnIh"
elif [ "$VOICE_LANGUAGE" = "en" ]; then
    export DIFY_API_KEY="app-Kw3QqjIXE6xtwjsGVxuvIZU8"
fi
```

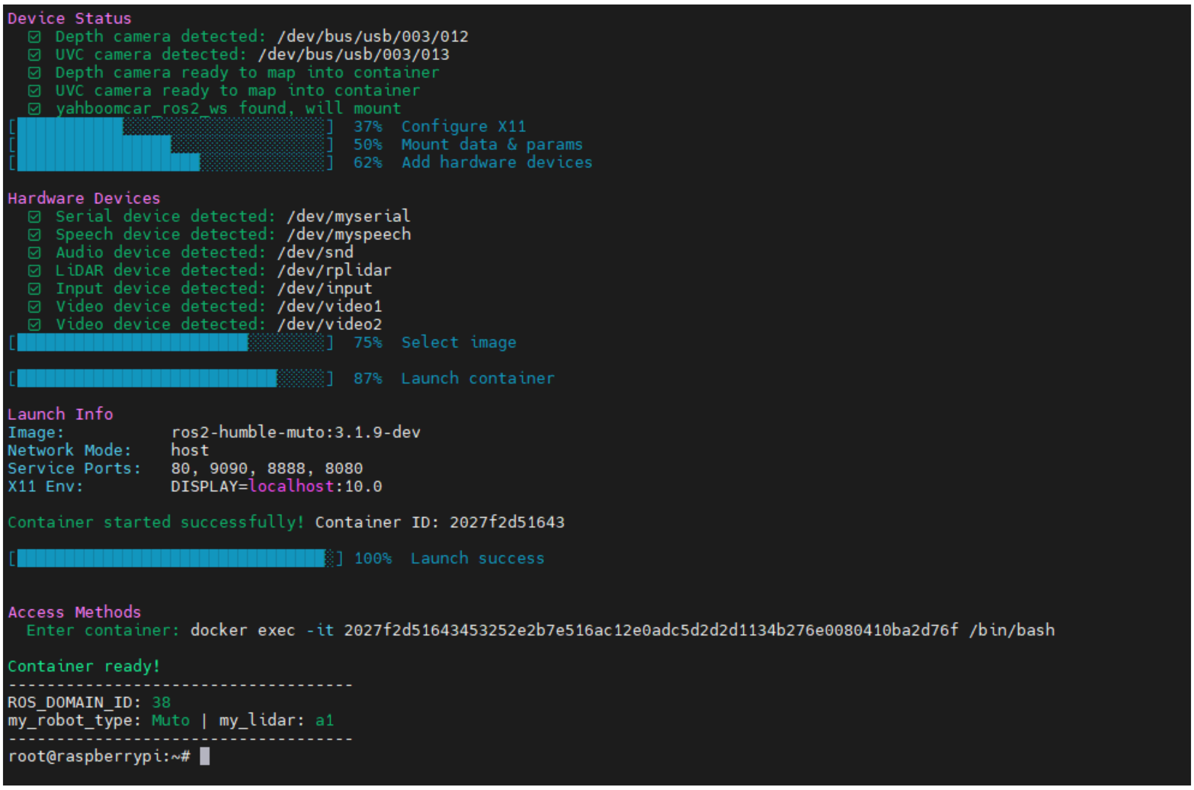






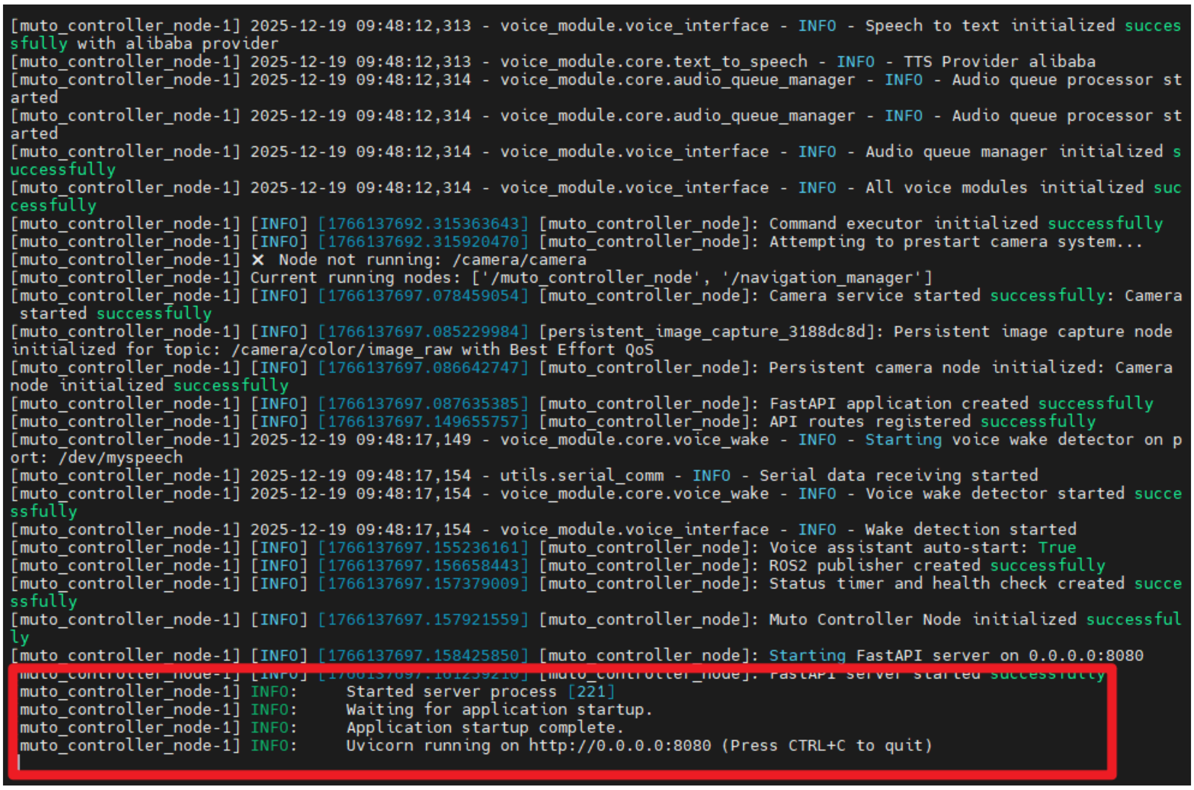
This lesson requires MutoRS to be powered on, as described in the previous chapters. Content: Starting the large model terminal in Docker





Starting the large model backend using the command

```
./rebuild_and_launch.sh
```



```
[muto_controller_node-1] 2025-12-19 09:48:12,313 - voice_module.voice_interface - INFO - Speech to text initialized succes
sfully with alibaba provider
[muto_controller_node-1] 2025-12-19 09:48:12,313 - voice_module.core.text_to_speech - INFO - TTS Provider alibaba
[muto_controller_node-1] 2025-12-19 09:48:12,314 - voice_module.core.audio_queue_manager - INFO - Audio queue processor st
arted
[muto_controller_node-1] 2025-12-19 09:48:12,314 - voice_module.core.audio_queue_manager - INFO - Audio queue processor st
arted
[muto_controller_node-1] 2025-12-19 09:48:12,314 - voice_module.voice_interface - INFO - Audio queue manager initialized s
uccessfully
[muto_controller_node-1] 2025-12-19 09:48:12,314 - voice_module.voice_interface - INFO - All voice modules initialized suc
cessfully
[muto_controller_node-1] [INFO] [1766137692.315363643] [muto_controller_node]: Command executor initialized successfully
[muto_controller_node-1] [INFO] [1766137692.315920470] [muto_controller_node]: Attempting to prestart camera system...
[muto_controller_node-1] ✕ Node not running: /camera/camera
[muto_controller_node-1] Current running nodes: ['/muto_controller_node', '/navigation_manager']
[muto_controller_node-1] [INFO] [1766137697.078459054] [muto_controller_node]: Camera service started successfully: Camera
 started successfully
[muto_controller_node-1] [INFO] [1766137697.085229984] [persistent_image_capture_3188dc8d]: Persistent image capture node
initialized for topic: /camera/color/image_raw with Best Effort QoS
[muto_controller_node-1] [INFO] [1766137697.086642747] [muto_controller_node]: Persistent camera node initialized: Camera
node initialized successfully
[muto_controller_node-1] [INFO] [1766137697.087635385] [muto_controller_node]: FastAPI application created successfully
[muto_controller_node-1] [INFO] [1766137697.149655757] [muto_controller_node]: API routes registered successfully
[muto_controller_node-1] 2025-12-19 09:48:17,149 - voice_module.core.voice_wake - INFO - Starting voice wake detector on p
ort: /dev/myspeech
[muto_controller_node-1] 2025-12-19 09:48:17,154 - utils.serial_comm - INFO - Serial data receiving started
[muto_controller_node-1] 2025-12-19 09:48:17,154 - voice_module.core.voice_wake - INFO - Voice wake detector started succe
ssfully
[muto_controller_node-1] 2025-12-19 09:48:17,154 - voice_module.voice_interface - INFO - Wake detection started
[muto_controller_node-1] [INFO] [1766137697.155236161] [muto_controller_node]: Voice assistant auto-start: True
[muto_controller_node-1] [INFO] [1766137697.156658443] [muto_controller_node]: ROS2 publisher created successfully
[muto_controller_node-1] [INFO] [1766137697.157379009] [muto_controller_node]: Status timer and health check created succe
ssfully
[muto_controller_node-1] [INFO] [1766137697.157921559] [muto_controller_node]: Muto Controller Node initialized successful
ly
[muto_controller_node-1] [INFO] [1766137697.158425850] [muto_controller_node]: Starting FastAPI server on 0.0.0.0:8080
muto_controller_node-1] INFO:     Started server process [221]
muto_controller_node-1] INFO:     Waiting for application startup.
muto_controller_node-1] INFO:     Application startup complete.
muto_controller_node-1] INFO:     Uvicorn running on http://0.0.0.0:8080 (Press CTRL+C to quit)
```

Getting Started

In this section, we'll combine navigation and line-following functionality. Therefore, in addition to the terminal program for the large model and Dify, we also need to open the terminal programs for navigation and line-following simultaneously.

Start the radar node:

```
ros2 launch yahboomcar_nav laser_bringup_launch.py
```

Start the radar odometry node:

```
ros2 launch rf2o_laser_odometry rf2o_laser_odometry.launch.py
```

Start the navigation node:

```
ros2 launch hexapod_nav hexapod_navigation_mppi.launch.py autostart:=true
map:=/root/yahboomcar_ros2_ws/maps/test_map.yaml
```

Please note that a pre-built map file (*.yaml and *.pgm; here we are using the map /root/yahboomcar_ros2_ws/maps/test_map) is required.

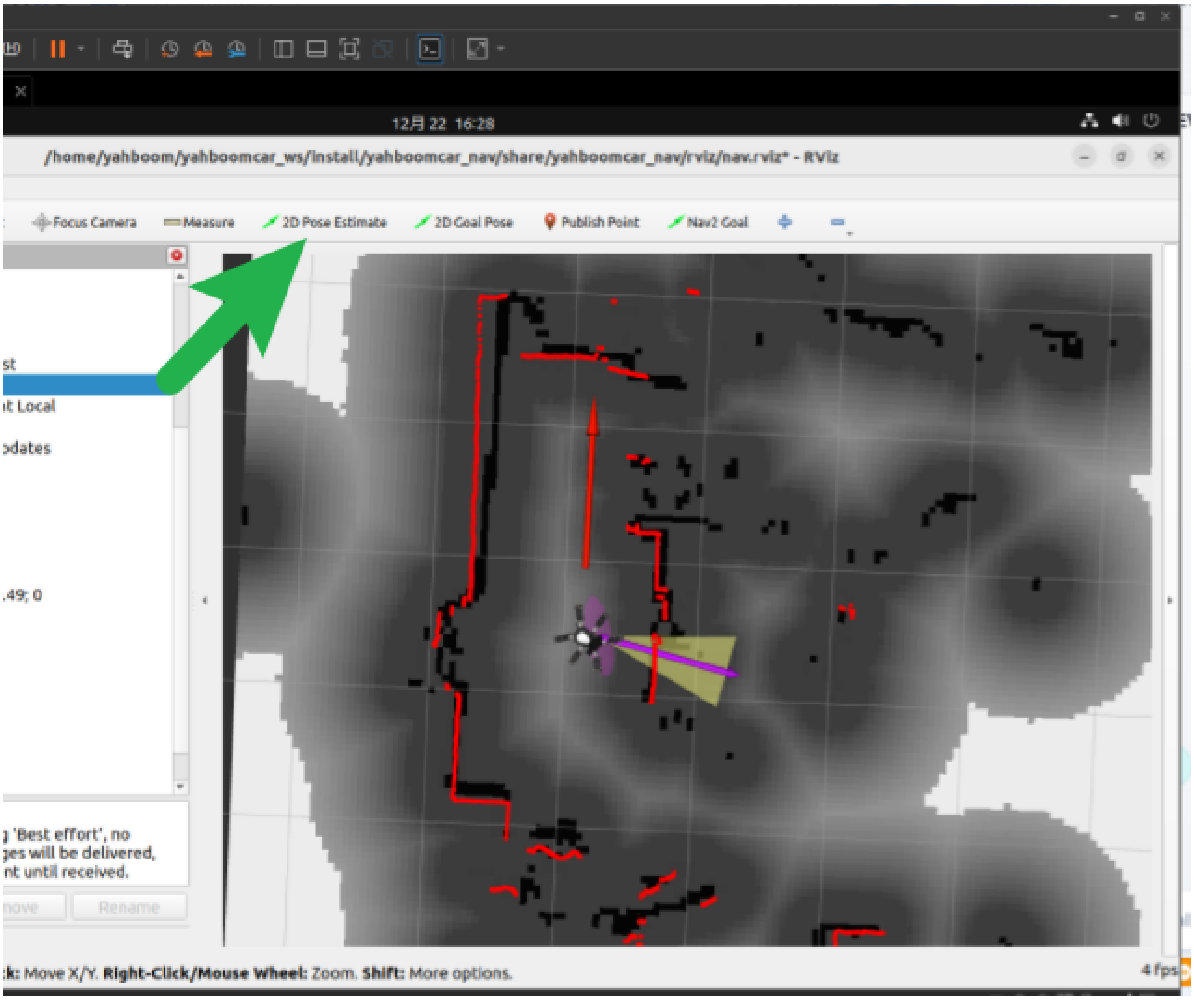
Finally, we can start the Rviz2 node to easily observe the navigation process.

This step is recommended to be performed in a VMware virtual machine! **

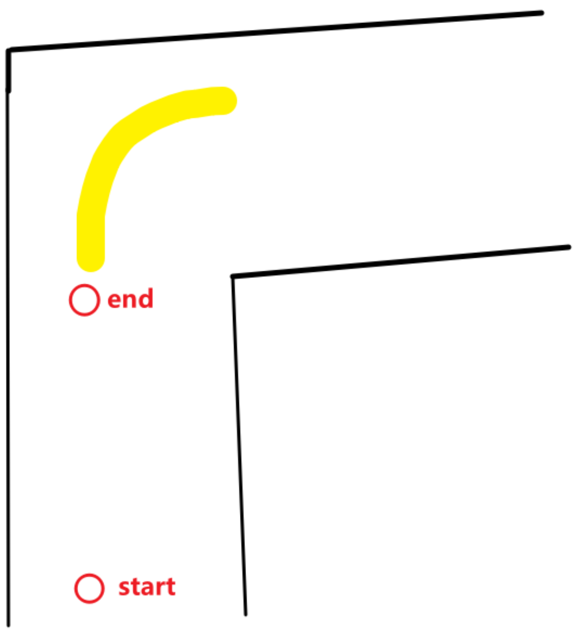
```
ros2 launch yahboomcar_nav display_nav_launch.py
```

It is crucial to change the selected area in the image to Transient Local here; otherwise, the map may not load.

After selecting it, we use the 2D Pose Estimate tool to set the initial position.



The scene where Muto is located is shown below.



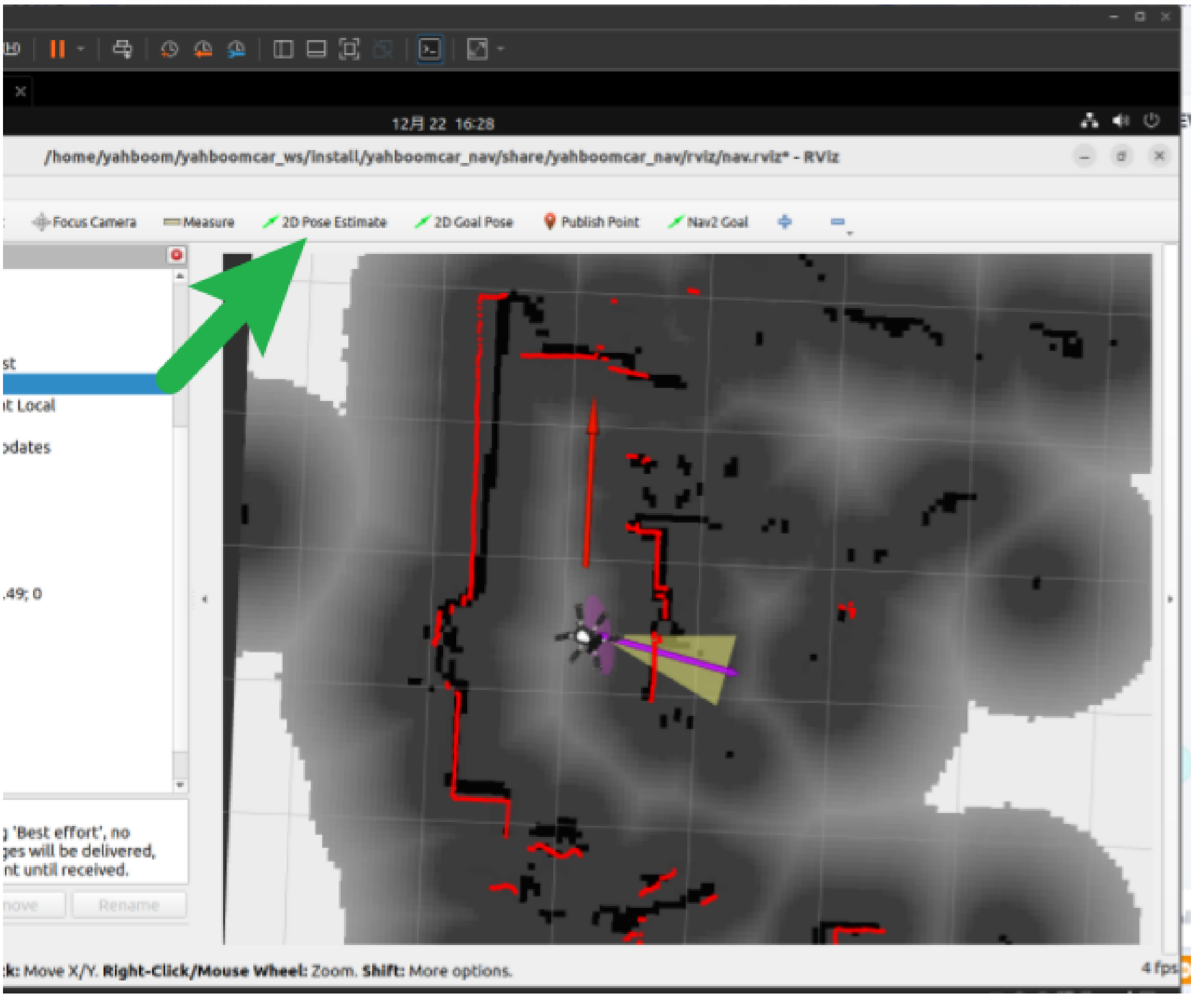
The yellow area represents the route to be followed. The yellow line, and the red circles "end" and "start" represent two pre-saved navigation points.

Our task is: "Let Muto navigate from any position to 'end,' then begin following the yellow line. After traversing the line, observe the surrounding environment, and finally navigate to the 'start' point."

Therefore, before starting, we need to record the HSV value of the yellow line (refer to previous chapters), and then record the two navigation points "start" and "end."

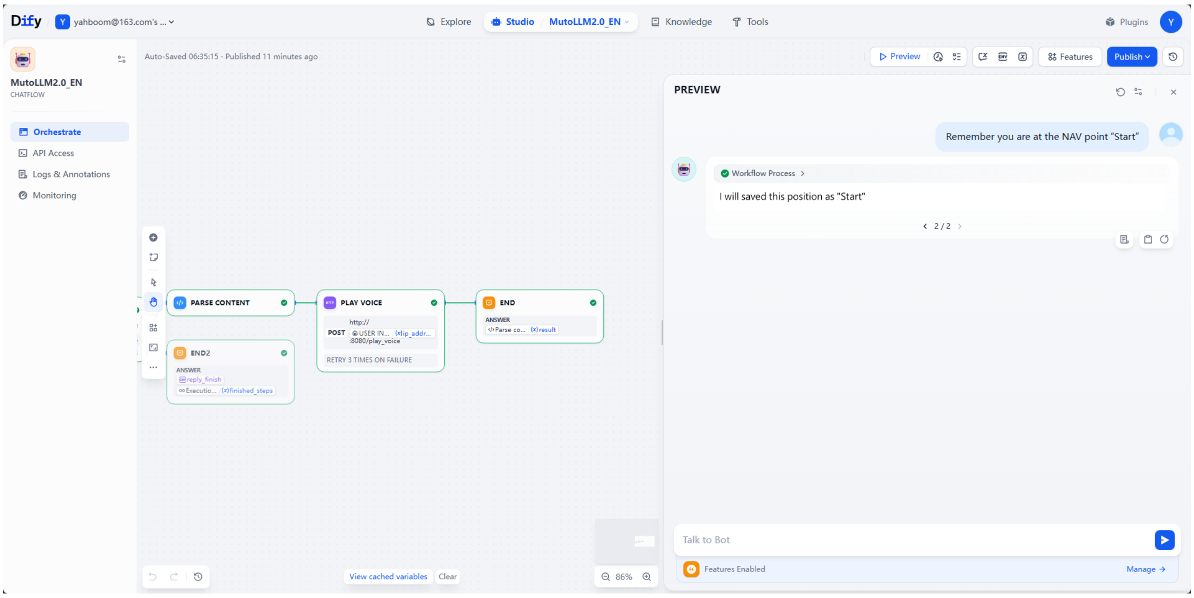
We'll first place Muto at the 'start' position.

Important! Before saving the navigation point, you need to set the initial position in Rviz using the 2D Pose Estimate tool.



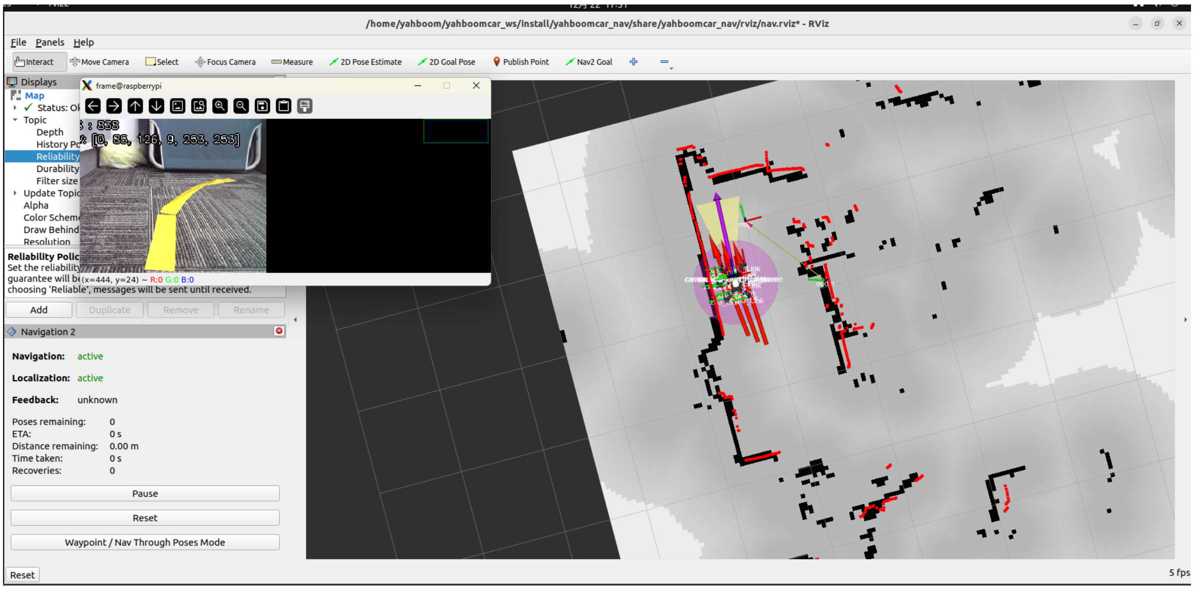
Then open the Divite terminal and type:

```
Remember that the current position is the navigation point "start"
```



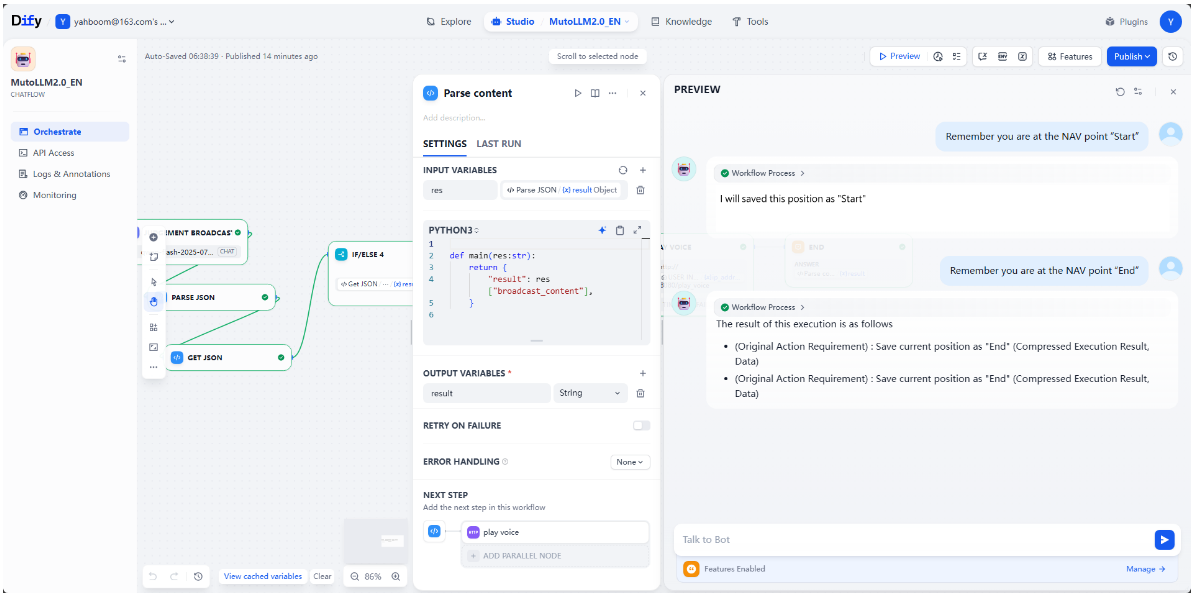
Next, manually move Muto to the starting point of the line, "end".

After moving, it's best to recalibrate the position using the RViz tool to minimize coordinate errors.



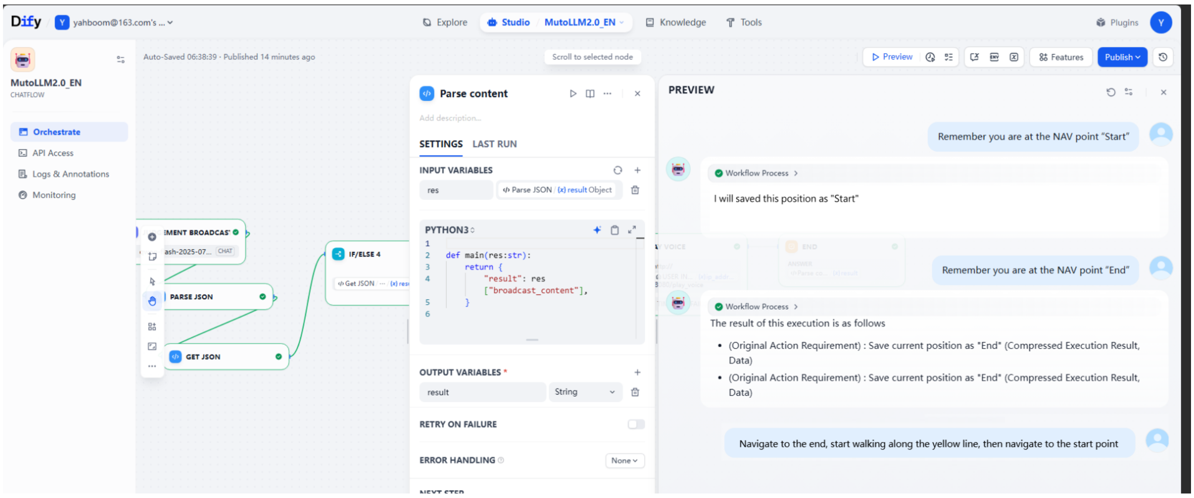
After moving, enter the following in the Dify terminal:

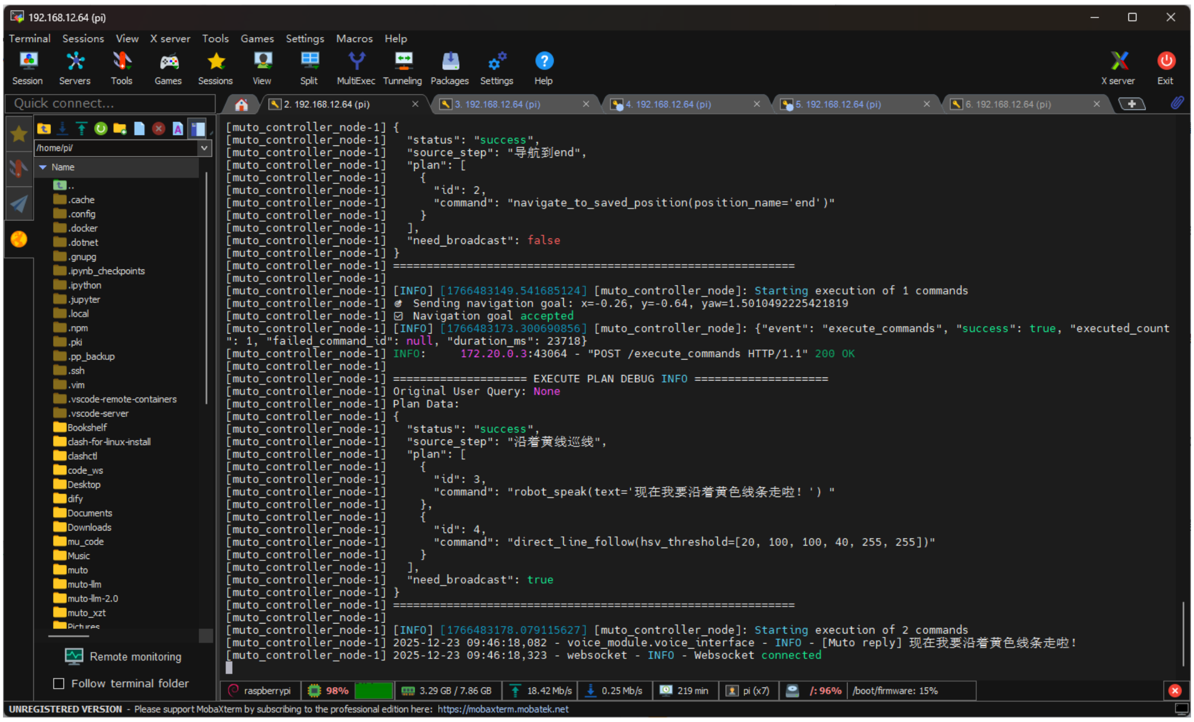
```
Remember that the current position is the navigation point "end"
```



Next, we move Muto to any position, then reconfirm the initial position using RViz, and then enter the following command in the Dify page:

```
Navigate to end, and then start following the yellow line. After completing the
line, observe the surrounding environment, and then navigate back to the start
point.
```




192.168.12.64 (pi)
[muto_controller_node-1] {
[muto_controller_node-1]   "status": "success",
[muto_controller_node-1]   "source_step": "导航到end",
[muto_controller_node-1]   "plan": [
[muto_controller_node-1]   {
[muto_controller_node-1]   "id": 2,
[muto_controller_node-1]   "command": "navigate_to_saved_position(position_name='end')"
[muto_controller_node-1]   }
[muto_controller_node-1]   ],
[muto_controller_node-1]   "need_broadcast": false
[muto_controller_node-1] }
[muto_controller_node-1] ==
[muto_controller_node-1]
[muto_controller_node-1] [INFO] [1766483149.541685124] [muto_controller_node]: Starting execution of 1 commands
[muto_controller_node-1] Sending navigation goal: x=-0.26, y=-0.64, yaw=1.5010492225421819
[muto_controller_node-1] ☑ Navigation goal accepted
[muto_controller_node-1] [INFO] [1766483173.300690856] [muto_controller_node]: {"event": "execute_commands", "success": true, "executed_count": 1, "failed_command_id": null, "duration_ms": 23718}
[muto_controller_node-1] INFO: 172.20.0.3:43064 - "POST /execute_commands HTTP/1.1" 200 OK
[muto_controller_node-1]
[muto_controller_node-1] ==================== EXECUTE PLAN DEBUG INFO ====================
[muto_controller_node-1] Original User Query: None
[muto_controller_node-1] Plan Data:
[muto_controller_node-1] {
[muto_controller_node-1]   "status": "success",
[muto_controller_node-1]   "source_step": "沿着黄线巡线",
[muto_controller_node-1]   "plan": [
[muto_controller_node-1]   {
[muto_controller_node-1]   "id": 3,
[muto_controller_node-1]   "command": "robot_speak(text='现在我要沿着黄色线条走啦！')"
[muto_controller_node-1]   },
[muto_controller_node-1]   {
[muto_controller_node-1]   "id": 4,
[muto_controller_node-1]   "command": "direct_line_follow(hsv_threshold=[20, 100, 100, 40, 255, 255])"
[muto_controller_node-1]   }
[muto_controller_node-1]   ],
[muto_controller_node-1]   "need_broadcast": true
[muto_controller_node-1] }
[muto_controller_node-1] ==
[muto_controller_node-1]
[muto_controller_node-1] [INFO] [1766483178.079115627] [muto_controller_node]: Starting execution of 2 commands
[muto_controller_node-1] 2025-12-23 09:46:18,082 - voice_module.voice_interface - INFO - [Muto reply] 现在我要沿着黄色线条走啦！
[muto_controller_node-1] 2025-12-23 09:46:18,323 - websocket - INFO - Websocket connected
raspberrypi 98% 3.29 GB / 7.86 GB 18.42 Mb/s 0.25 Mb/s 219 min pi (x7) /: 96% /boot/firmware: 15%
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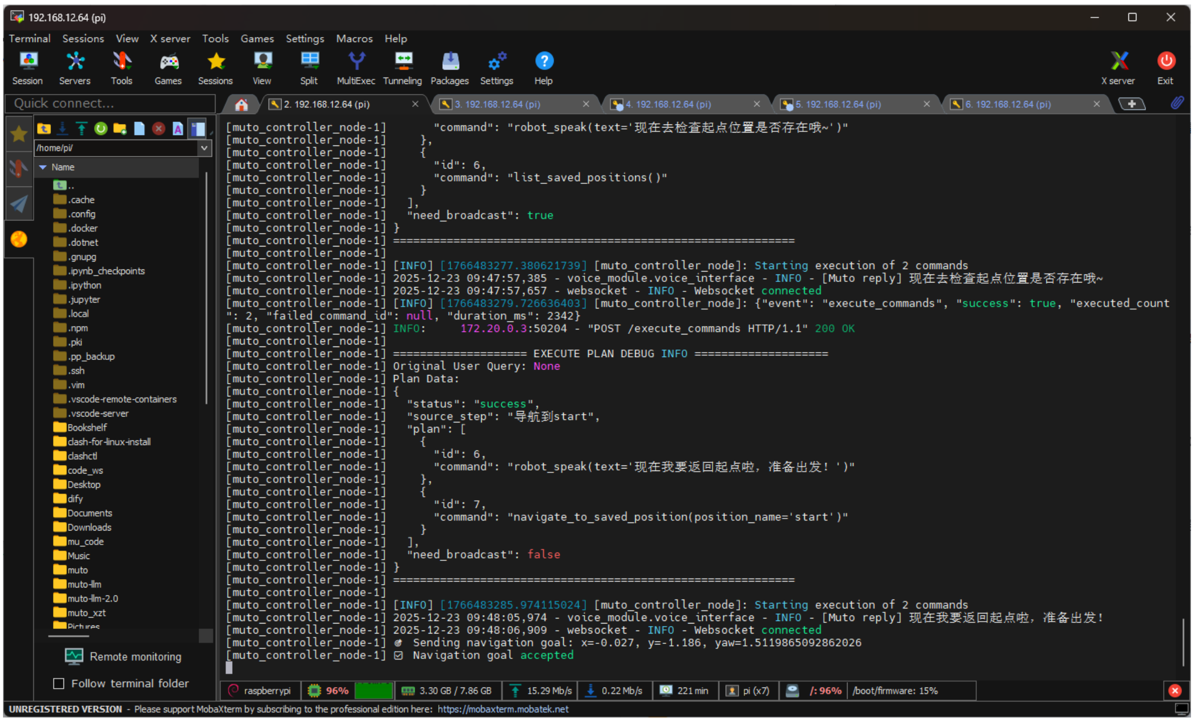
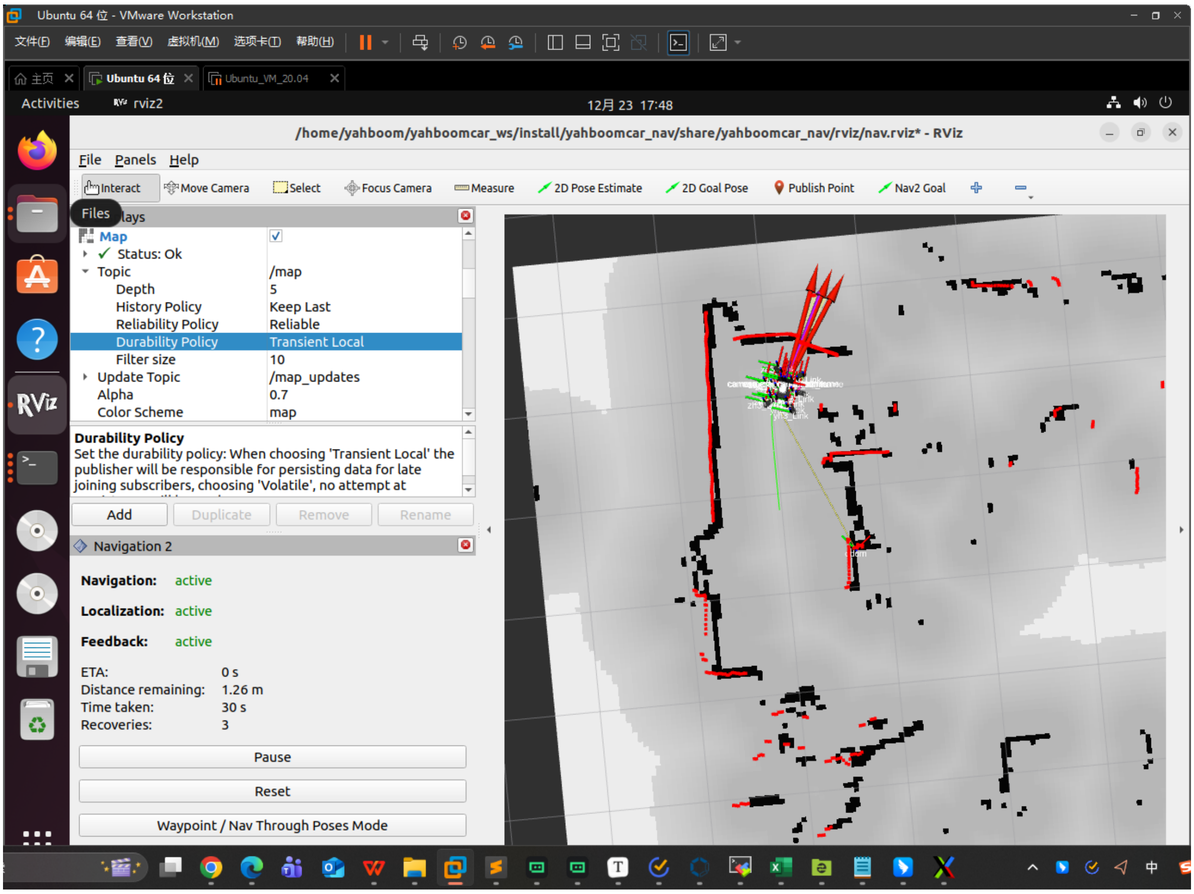

192.168.12.64 (pi)
[muto_controller_node-1]   "command": "robot_speak(text='现在去检查起点位置是否存在哦~')"
[muto_controller_node-1]   },
[muto_controller_node-1]   {
[muto_controller_node-1]   "id": 6,
[muto_controller_node-1]   "command": "list_saved_positions()"
[muto_controller_node-1]   }
[muto_controller_node-1]   ],
[muto_controller_node-1]   "need_broadcast": true
[muto_controller_node-1] }
[muto_controller_node-1] ==
[muto_controller_node-1]
[muto_controller_node-1] [INFO] [1766483277.380621739] [muto_controller_node]: Starting execution of 2 commands
[muto_controller_node-1] 2025-12-23 09:47:57,385 - voice_module.voice_interface - INFO - [Muto reply] 现在去检查起点位置是否存在哦~
[muto_controller_node-1] 2025-12-23 09:47:57,657 - websocket - INFO - Websocket connected
[muto_controller_node-1] [INFO] [1766483279.726636403] [muto_controller_node]: {"event": "execute_commands", "success": true, "executed_count": 2, "failed_command_id": null, "duration_ms": 2342}
[muto_controller_node-1] INFO: 172.20.0.3:50204 - "POST /execute_commands HTTP/1.1" 200 OK
[muto_controller_node-1]
[muto_controller_node-1] ==================== EXECUTE PLAN DEBUG INFO ====================
[muto_controller_node-1] Original User Query: None
[muto_controller_node-1] Plan Data:
[muto_controller_node-1] {
[muto_controller_node-1]   "status": "success",
[muto_controller_node-1]   "source_step": "导航到start",
[muto_controller_node-1]   "plan": [
[muto_controller_node-1]   {
[muto_controller_node-1]   "id": 6,
[muto_controller_node-1]   "command": "robot_speak(text='现在我要返回起点啦，准备出发！')"
[muto_controller_node-1]   },
[muto_controller_node-1]   {
[muto_controller_node-1]   "id": 7,
[muto_controller_node-1]   "command": "navigate_to_saved_position(position_name='start')"
[muto_controller_node-1]   }
[muto_controller_node-1]   ],
[muto_controller_node-1]   "need_broadcast": false
[muto_controller_node-1] }
[muto_controller_node-1] ==
[muto_controller_node-1]
[muto_controller_node-1] [INFO] [1766483285.974115024] [muto_controller_node]: Starting execution of 2 commands
[muto_controller_node-1] 2025-12-23 09:48:05,974 - voice_module.voice_interface - INFO - [Muto reply] 现在我要返回起点啦，准备出发！
[muto_controller_node-1] 2025-12-23 09:48:06,909 - websocket - INFO - Websocket connected
[muto_controller_node-1] Sending navigation goal: x=-0.027, y=-1.186, yaw=1.5119865092862026
[muto_controller_node-1] ☑ Navigation goal accepted
raspberrypi 96% 3.30 GB / 7.86 GB 15.29 Mb/s 0.22 Mb/s 221 min pi (x7) /: 96% /boot/firmware: 15%
UNREGISTERED VERSION - Please support MobaXterm by subscribing to the professional edition here: https://mobaxterm.mobatek.net



Finally, you can see that Muto has returned to the starting point.